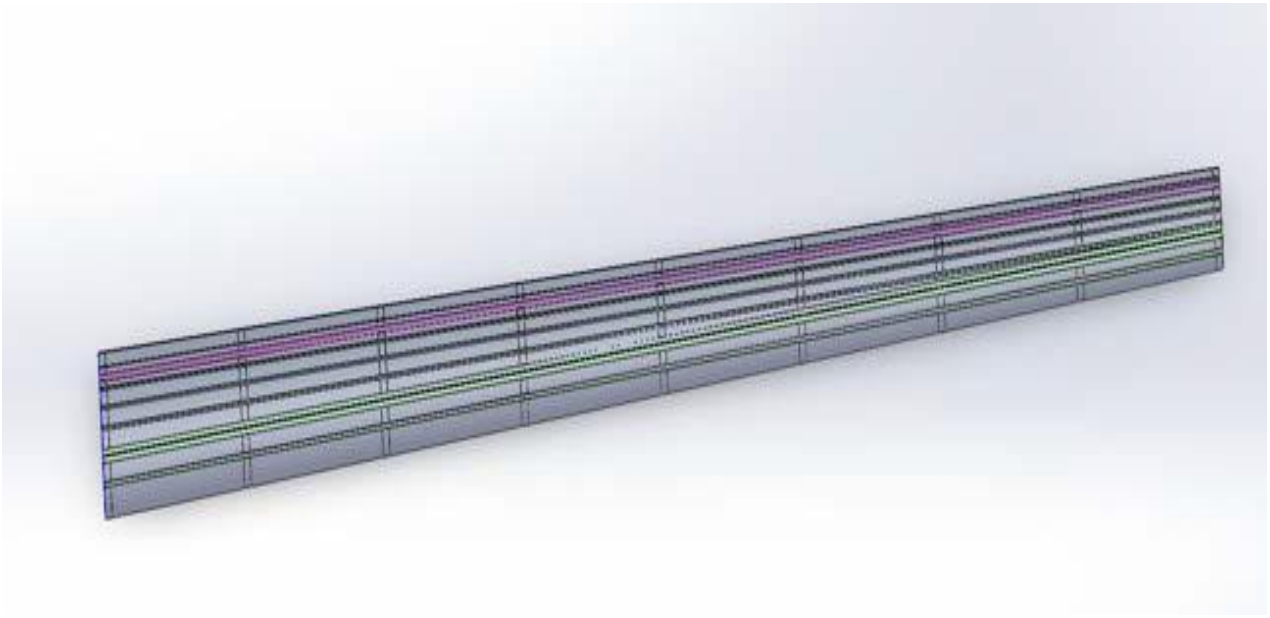
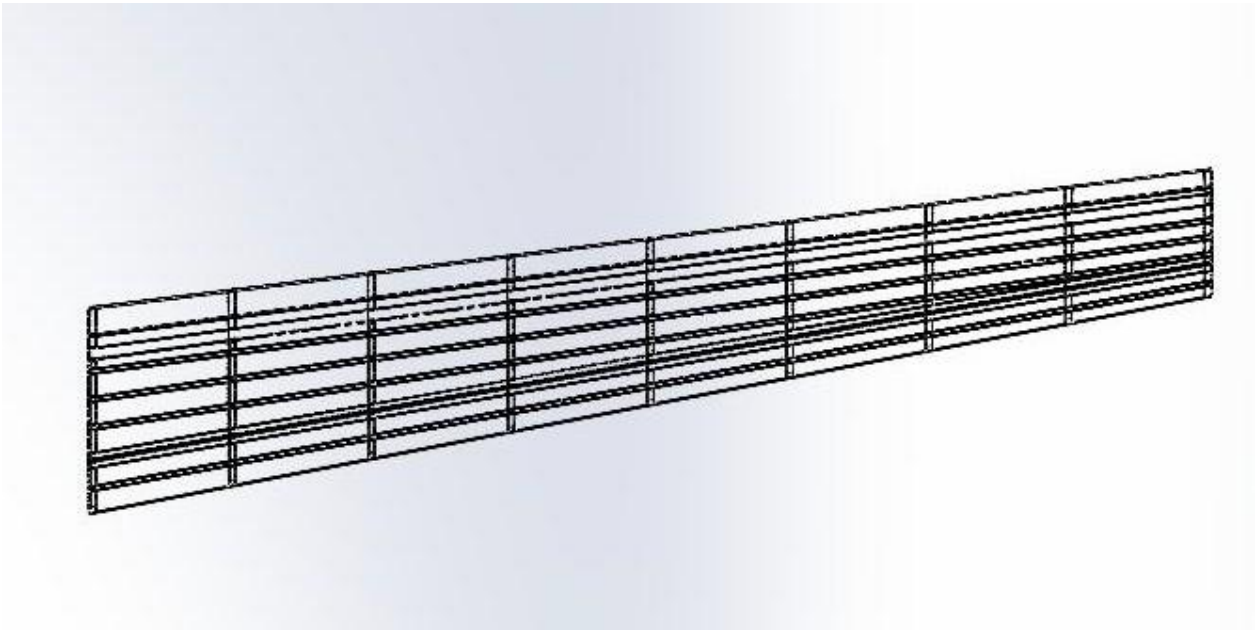
**A PROJECT REPORT ON**

# FINITE ELEMENT ANALYSIS OF A UAV WING USING THE NACA 4412 AIRFOIL





# ABSTRACT

This project focuses on the structural analysis of a NACA 4412 UAV wing using ANSYS Mechanical. A half-span wing model featuring a 2 mm carbon-fibre skin supported by an internal rib-and-spar structure was developed and analysed using Static Structural analysis. A fixed support was applied at the wing root, with an aerodynamic pressure load of 84.1 Pa acting on the upper skin. The analysis evaluated total deformation, von Mises stress, and equivalent elastic strain. The results showed a maximum deformation of 0.867 mm, maximum equivalent stress of 16.811 MPa, and maximum elastic strain of $1.2012 \times 10^{-4}$ mm/mm. Based on a structural steel yield strength of 250 MPa, the calculated factor of safety was approximately 14.9. The results indicate that the designed wing structure can withstand the applied loading with minimal deformation and stresses within the assumed material yield limit

# INTRODUCTION

## Background

A UAV wing consists of an aerodynamic skin supported by internal ribs and spars. These components maintain the wing shape and provide structural strength to withstand aerodynamic loads. This project models a realistic wing structure for structural analysis.

## Objectives

- Model a NACA 4412 UAV wing with skin, ribs and spars.
- Assign carbon-fibre skin and structural-steel internal components.
- Perform meshing and static structural analysis in ANSYS.
- Evaluate deformation, von Mises stress and elastic strain.
- Determine the factor of safety under the applied aerodynamic pressure.

## Scope

The project focuses on linear static structural analysis of a half-span UAV wing under aerodynamic pressure. Composite ply-level analysis, vibration, fatigue, buckling and aeroelastic analysis are excluded and considered future work.

# GEOMETRY AND MODEL DESCRIPTION

## Wing Geometry

The analysed structure is one half of the wing (root to tip), modelled explicitly with an outer aerodynamic skin and an internal rib-and-spar structure. The geometry is consistent with the baseline wing used throughout this project:

| Parameter | Value |
|---|---|
| Half-span (root to tip) analysed | 1000 mm (1.000 m) |
| Full wing span (both halves) | 2000 mm (2.000 m) |
| Root Chord | 180 mm |
| Tip Chord | 110 mm |
| Taper Ratio (Tip/Root) | 0.611 — consistent with the XFLR5 baseline wing |
| Airfoil Section | NACA 4412 |

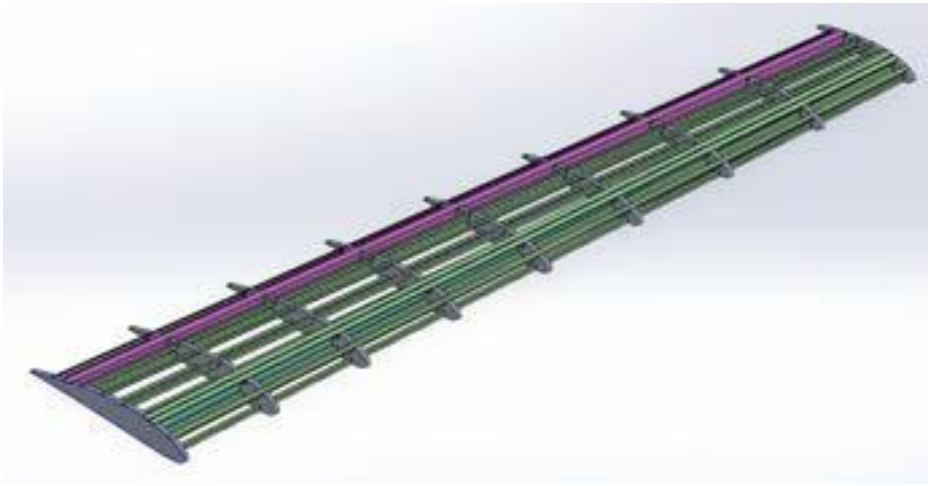


*Figure 2.1: Half-wing internal structure — skin, ribs, and spars (SolidWorks model)*

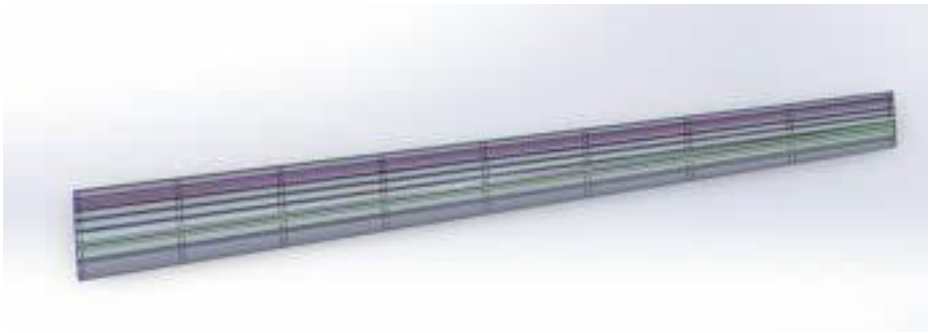


*Figure 2.2: Half-wing assembly — shaded view showing rib spacing along the span*



*Figure 2.3: Wireframe view showing the full internal rib/spar layout*

# MATERIAL PROPERTIES

## Skin - Carbon Fibre

The wing skin was assigned the Ansys “Carbon Fibre (290 GPa)” material, modelled as a 2 mm thick shell with Membrane and Bending stiffness behaviour.

| Property | Value |
|---|---|
| Density | 1800 kg/m³ |
| Young's Modulus — X (fibre direction) | $2.9 \times 10^{11}$ Pa (290 GPa) |
| Young's Modulus — Y (transverse) | $2.3 \times 10^{10}$ Pa (23 GPa) |
| Young's Modulus — Z (transverse) | $2.3 \times 10^{10}$ Pa (23 GPa) |
| Poisson's Ratio — XY | 0.20 |
| Poisson's Ratio — YZ | 0.40 |
| Poisson's Ratio — XZ | 0.20 |
| Shear Modulus — XY | $9.0 \times 10^{9}$ Pa |
| Shear Modulus — YZ | $8.2143 \times 10^{9}$ Pa |
| Shear Modulus — XZ | $9.0 \times 10^{9}$ Pa |

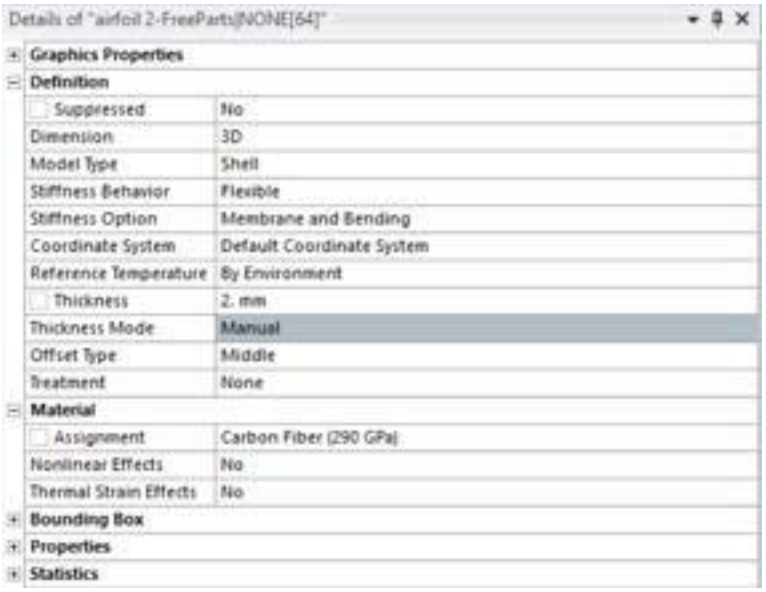



*Figure 3.1: Skin shell definition — 2 mm thickness, Carbon Fibre (290 GPa)*

### Ribs and Spars — Structural Steel

The internal ribs and spars were assigned the Ansys generic “Structural Steel” material, using the standard library default properties:

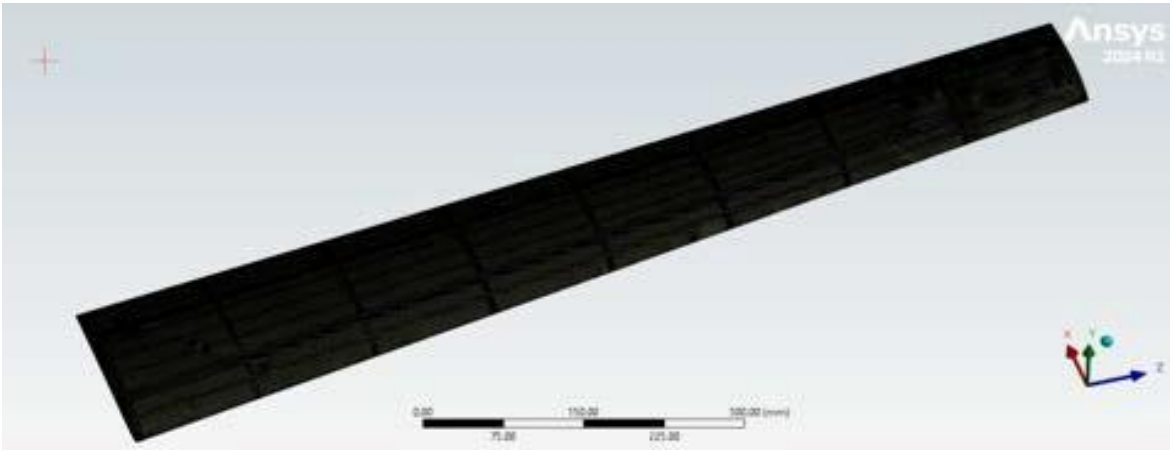
| Property | Typical Value (Ansys Structural Steel default) |
|---|---|
| Density | 7850 kg/m³ |
| Young's Modulus | 2.0 × 10¹¹ Pa (200 GPa) |
| Poisson's Ratio | 0.30 |
| Tensile Yield Strength | 250 MPa |
| Tensile Ultimate Strength | 460 MPa |

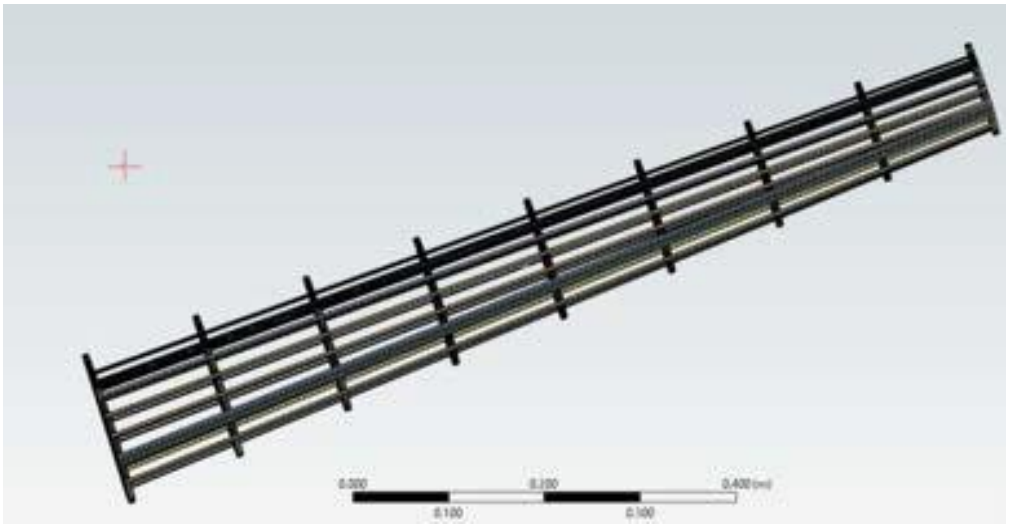
## MESHING

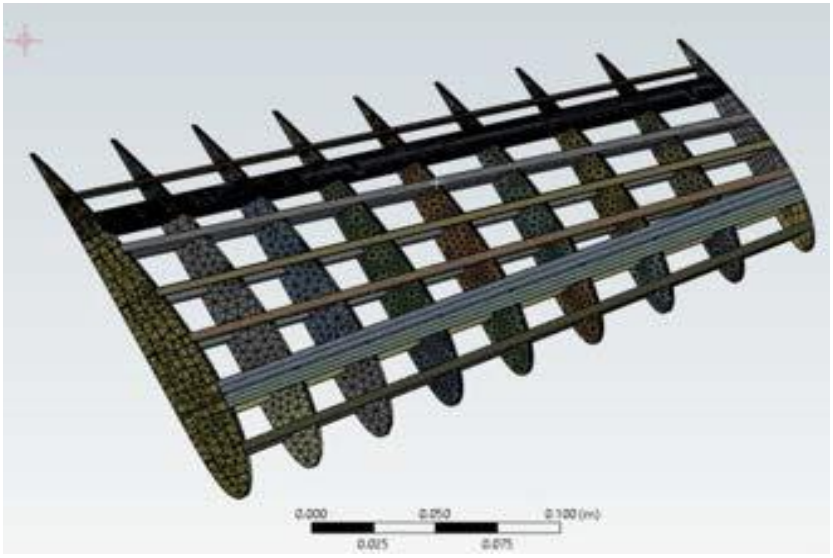
### Mesh Settings




*Figure 4.1: Mesh statistics and sizing settings*





The mesh was assessed using aspect ratio, skewness and element quality metrics. Local refinement was used to improve element distribution around structural features and intersections. A mesh-convergence study remains recommended to establish numerical mesh independence.

# BOUNDARY CONDITIONS

## Fixed Support

A Fixed Support was applied at the wing root, representing the root/symmetry boundary of the half-wing model (i.e. the plane at which the analysed half would connect to the aircraft centreline or the mirrored wing half).

## Pressure Load

A surface pressure of 8.41 × $10^{-5}$ MPa (84.1 Pa) was applied to the upper wing skin, shown as

load B in Figure 5.1.

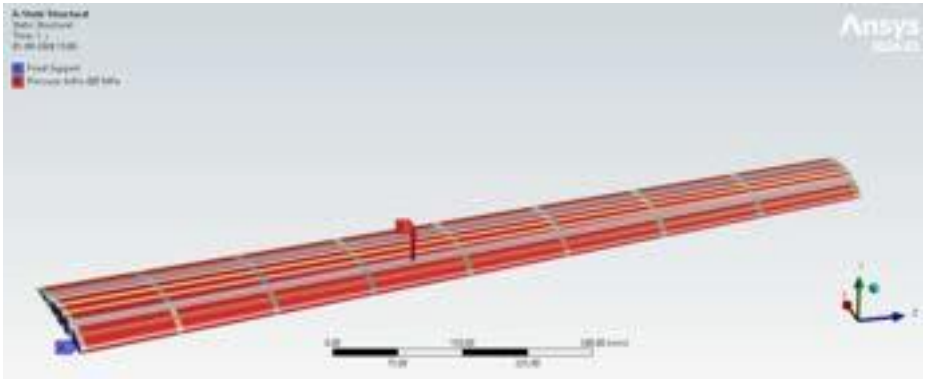


*Figure 5.1: Boundary conditions — Fixed Support (A) at root, Pressure (B) on upper skin*

From the earlier aerodynamic values:

$$L = \frac{1}{2}\rho V^2 S C_L$$

with $V = 10\text{m/s}$, $S = 0.290\text{m}^2$ and $C_L = 1.37$,

$$L \approx 24.4\ N$$

and

$$\frac{24.4}{0.290} \approx 84.1\ Pa$$

So **84.1 Pa is an equivalent average pressure based on the full-wing lift**.

But your structural model is a **half-wing**. Therefore, for a symmetric half-wing, the corresponding lift is:

$$L_{half} \approx 12.2\ N$$

and the half-wing planform area is approximately:

$$S_{half} = 0.145\ m^2$$

giving the same equivalent average pressure:

$$P_{avg} = \frac{12.2}{0.145} \approx 84.1\ Pa$$

So the number itself is consistent **as an equivalent uniform pressure**, but you should explicitly state that it is a simplification rather than an actual XFLR5 upper-surface pressure distribution.

# RESULTS

## Total Deformation

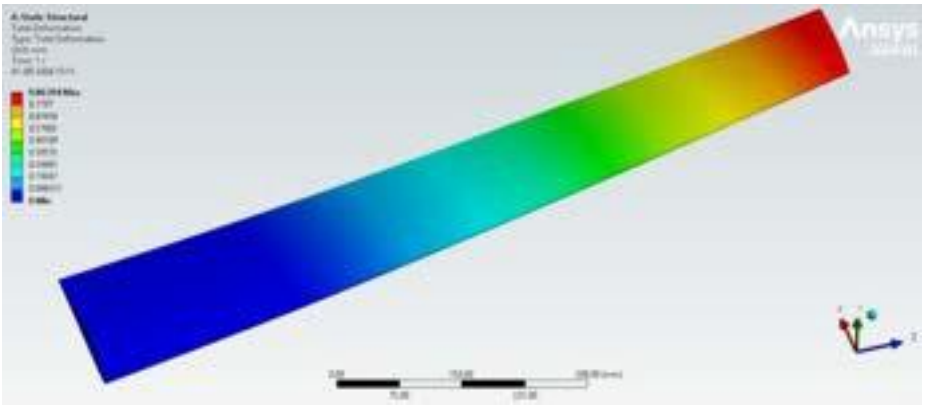


*Figure 6.1: Total deformation — maximum 0.867 mm at the wing tip*

The maximum total deformation of 0.867 mm occurs at the wing tip, increasing smoothly from zero at the fixed root, consistent with the expected cantilever bending behaviour of a wing structure under distributed upward pressure loading.

## Equivalent (von-Mises) Stress

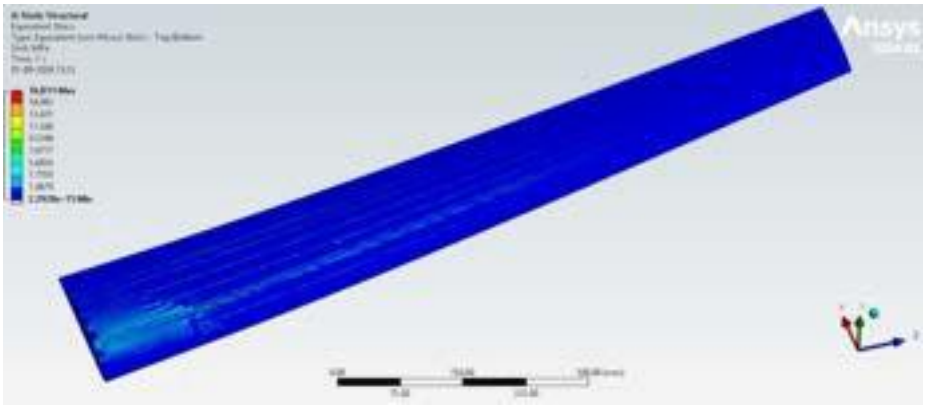


*Figure 6.2: Equivalent (von-Mises) stress — maximum 16.811 MPa*

The maximum von Mises stress of 16.811 MPa occurs in the internal rib-and-spar structure near the wing root, as shown in Figure 6.2. This is due to the higher bending moment and shear forces experienced at the wing root.

## Equivalent Elastic Strain

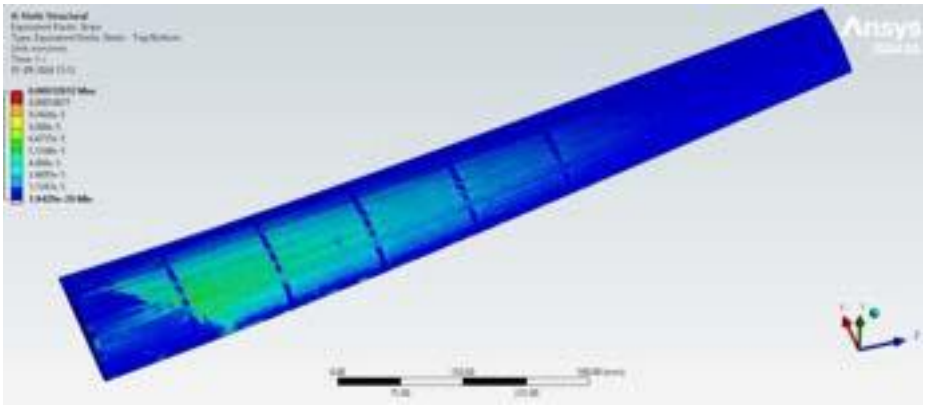


*Figure 6.3: Equivalent elastic strain — maximum 1.2012 × 10⁻⁴ mm/mm*

The calculated elastic strain remains within the linear-elastic material model used in the present analysis. A definitive strength assessment of the carbon-fibre skin requires directional composite strength allowables and an appropriate composite failure criterion.

## CONCLUSION

- Performed static structural analysis of a NACA 4412 half-span UAV wing with a carbon-fibre skin and internal rib-and-spar structure using ANSYS Mechanical.
- Defined orthotropic carbon-fibre and structural steel materials, and evaluated deformation, von Mises stress and elastic strain under aerodynamic pressure loading.
- Obtained a maximum deformation of 0.867 mm, peak stress of 16.811 MPa, and a yield-based factor of safety of 14.9, confirming structural strength under the applied load case.

## FUTURE SCOPE AND LIMITATIONS

- Perform composite failure analysis using directional strength properties and Tsai-Wu or maximum stress criteria.
- Verify fibre orientation and material coordinate alignment across the wing skin.
- Conduct mesh convergence studies to ensure result accuracy.
- Extend the analysis to modal, buckling and fatigue studies.
- Perform fluid–structure interaction (FSI) analysis using realistic CFD pressure distributions.